Hands-On Activity 10 | Explaining Visually to Influence with Data

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Course Code and Title : CPE031 / Visualizations and Data Analysis
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1. Objectives:

This activity aims to help students demonstrate how to influence and communicate effectively with data by identifying audience needs, crafting a clear message, and telling a story visually through charts.

2. Intended Learning Outcomes (ILOs):

By the end of this activity, students should be able to:

1. Analyze the audience to understand their level of data knowledge and needs.
2. Evaluate how well a message and visualization communicate insights.
3. Create a clear and meaningful data story using Python visualization tools.

3. Discussions:

Data storytelling goes beyond showing charts it's about influencing decisions and sparking understanding.

Know Your Audience: Decision-makers may prefer clear summaries, while analysts like detailed data.

Know Your Message: Focus on the why it matters part of the story.

Tell a Story: Use preattentive attributes color, size, and position to guide attention.

Structure Your Story: Like Aristotle's rhetorical triangle, combine logic (logos), emotion (pathos), and credibility (ethos).

4. Procedures:

Step 1. Import Libraries and Create Sample Data

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

# Sample sales dataset
data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May", "Jun"],
    "Sales": [200, 250, 180, 300, 280, 350],
    "Marketing_Spend": [50, 60, 45, 70, 65, 80]
}
df = pd.DataFrame(data)
df
```

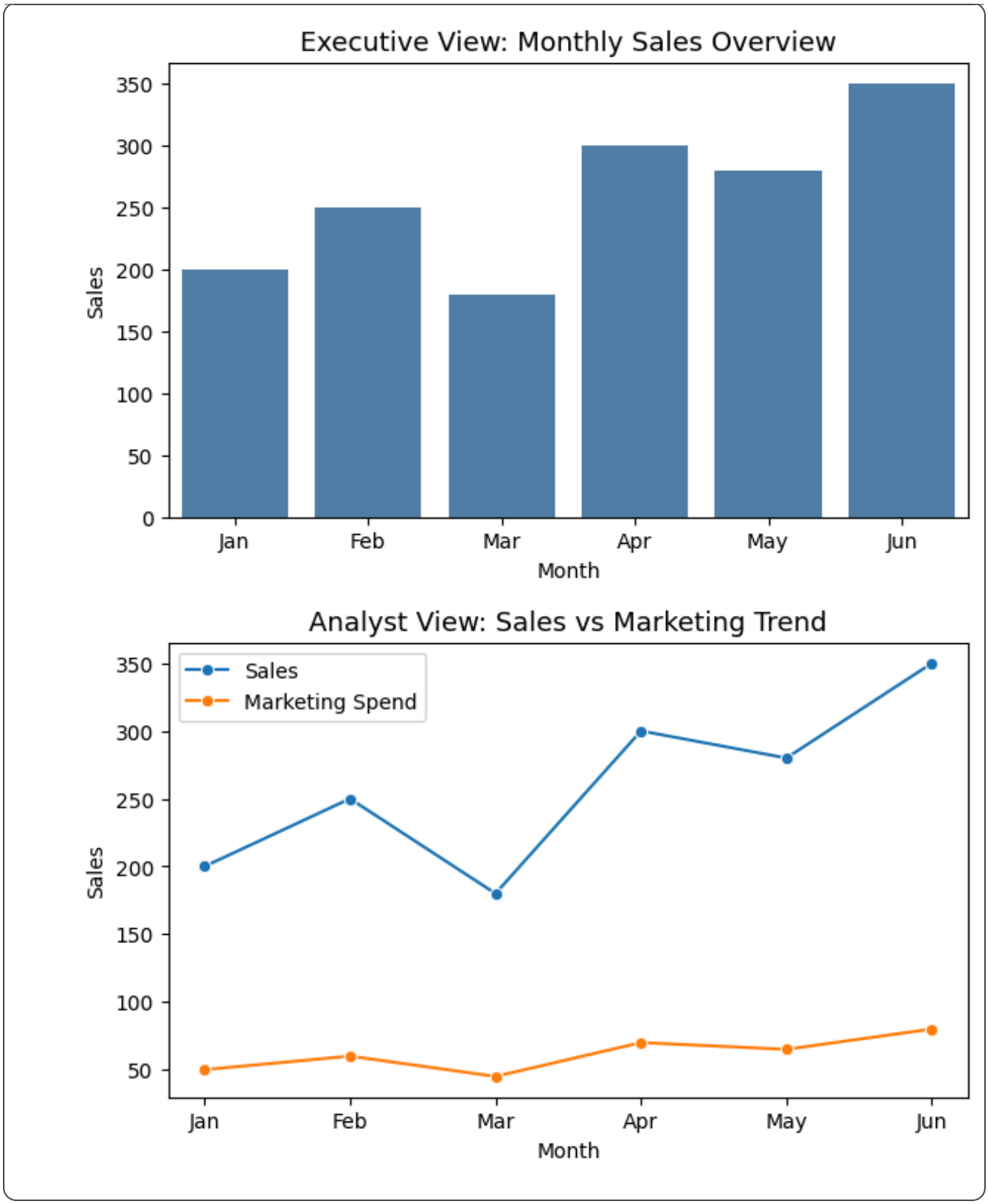
	Month	Sales	Marketing_Spend
0	Jan	200	50
1	Feb	250	60
2	Mar	180	45
3	Apr	300	70
4	May	280	65
5	Jun	350	80

Next steps: Generate code with df | New interactive sheet

Step 2. Know Your Audience — Compare Chart Styles

```
# Executive audience: prefer clarity (simple bar)
plt.figure(figsize=(7,4))
sns.barplot(x="Month", y="Sales", data=df, color="steelblue")
plt.title("Executive View: Monthly Sales Overview", fontsize=13)
plt.show()

# Analyst audience: prefer detailed trends (line chart)
plt.figure(figsize=(7,4))
sns.lineplot(x="Month", y="Sales", data=df, marker="o", label="Sales")
sns.lineplot(x="Month", y="Marketing_Spend", data=df, marker="o", label="Ma
plt.title("Analyst View: Sales vs Marketing Trend", fontsize=13)
plt.legend()
plt.show()
```

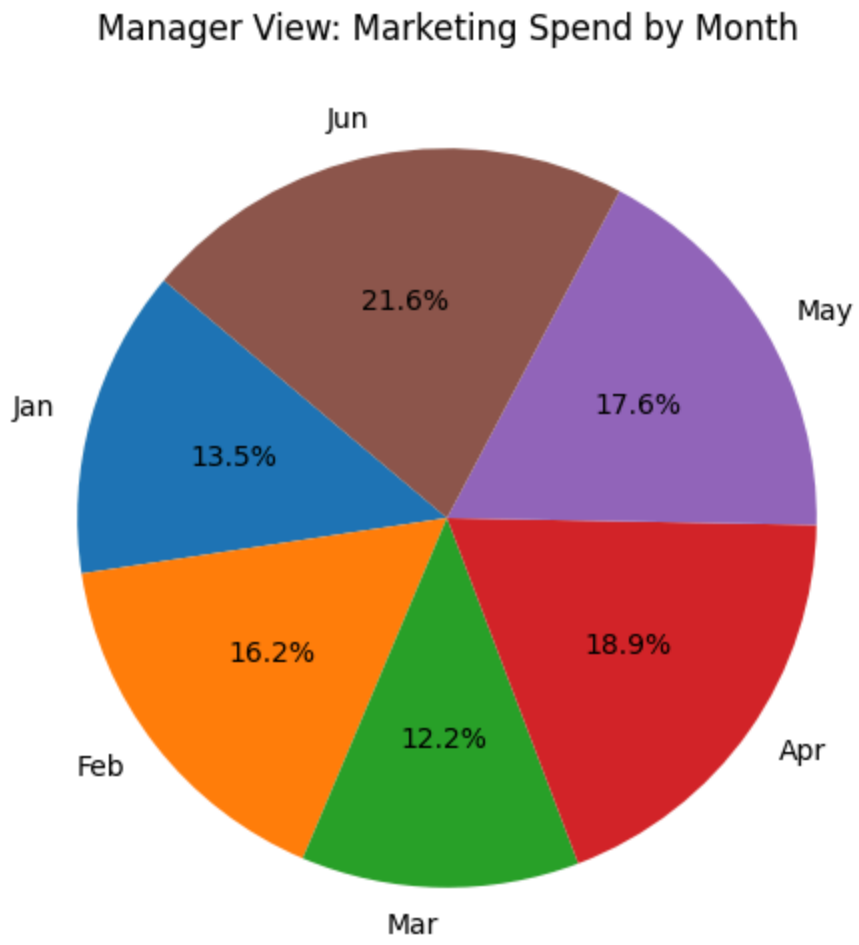


Task 1: Create one additional visualization suited for a different audience (e.g., a customer, investor, or teammate).

```
plt.figure(figsize=(6, 6))
plt.pie(
    df["Marketing_Spend"],
```

```
    labels=df["Month"],
    autopct='%1.1f%%',
    startangle=140
)

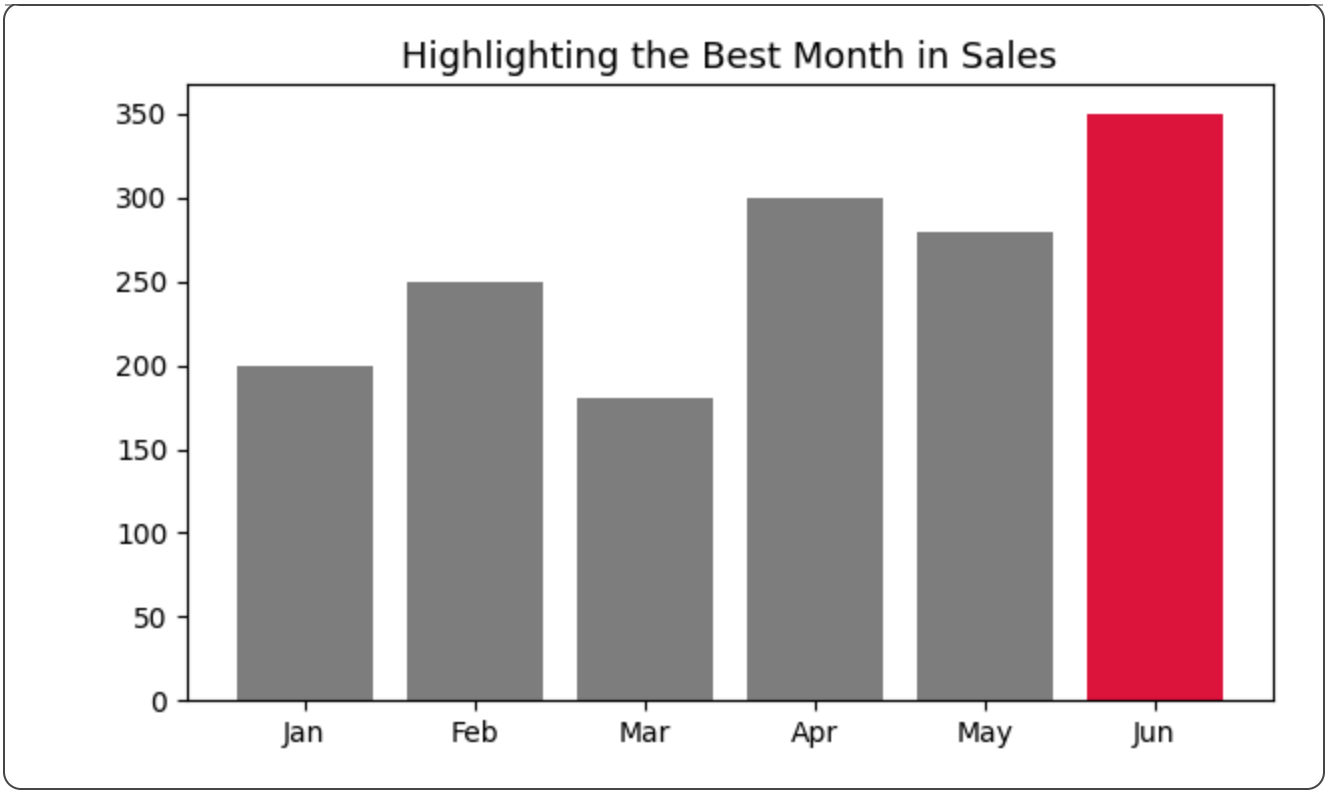
plt.title("Manager View: Marketing Spend by Month")
plt.show()
```



Step 3. Know Your Message — Highlight What Matters

```
# Highlight the highest sales month (preattentive color cue)
highlight = df["Sales"].idxmax()

colors = ["gray" if i != highlight else "crimson" for i in range(len(df))]
plt.figure(figsize=(7,4))
plt.bar(df["Month"], df["Sales"], color=colors)
plt.title("Highlighting the Best Month in Sales", fontsize=13)
plt.show()
```



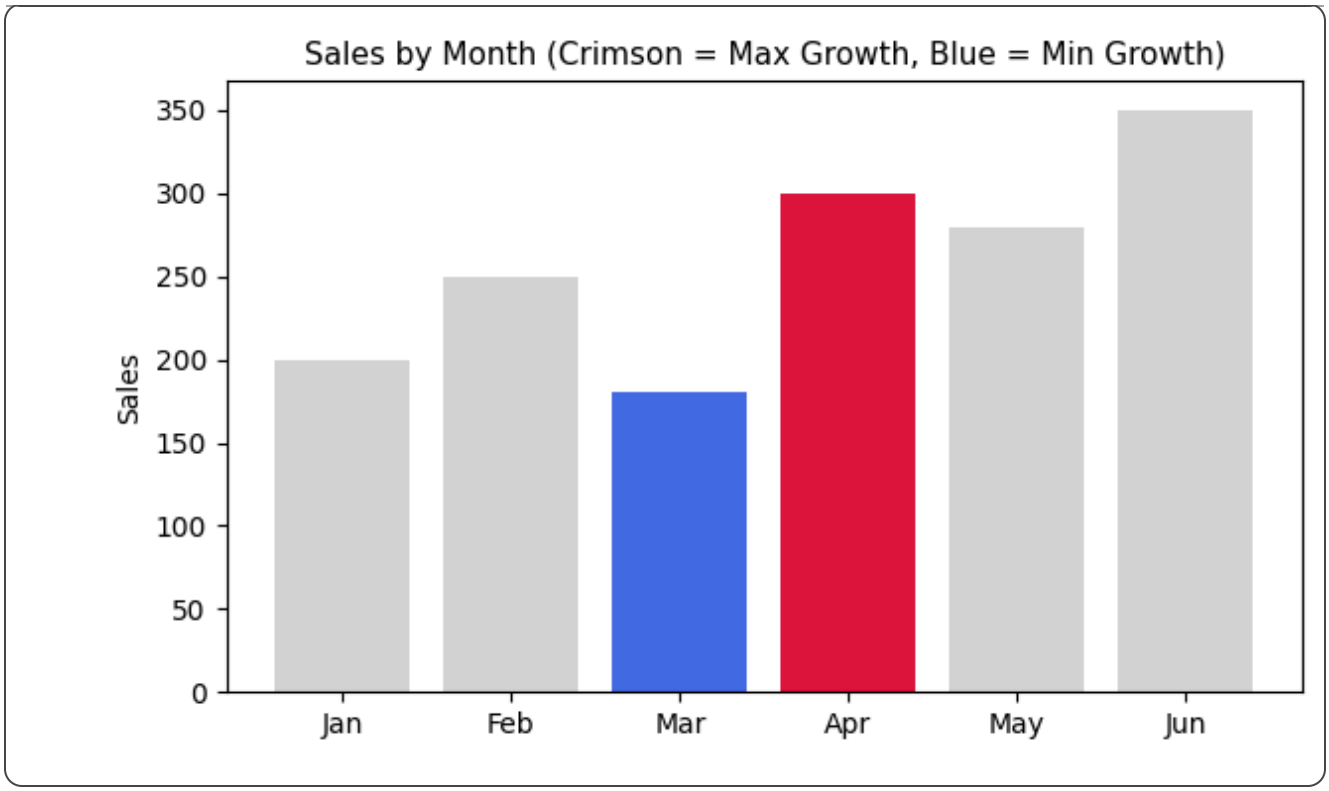
Task 2: Use color or annotations to emphasize a key insight (e.g., lowest point, sudden increase).

```
df["Growth"] = df["Sales"].diff()

max_growth = df["Growth"].idxmax()
min_growth = df["Growth"].idxmin()

colors = []
for i in range(len(df)):
    if i == max_growth:
        colors.append("crimson")
    elif i == min_growth:
        colors.append("royalblue")
    else:
        colors.append("lightgray")

plt.figure(figsize=(7, 4))
plt.bar(df["Month"], df["Sales"], color=colors)
plt.title("Sales by Month (Crimson = Max Growth, Blue = Min Growth)", fonts
plt.ylabel("Sales")
plt.show()
```

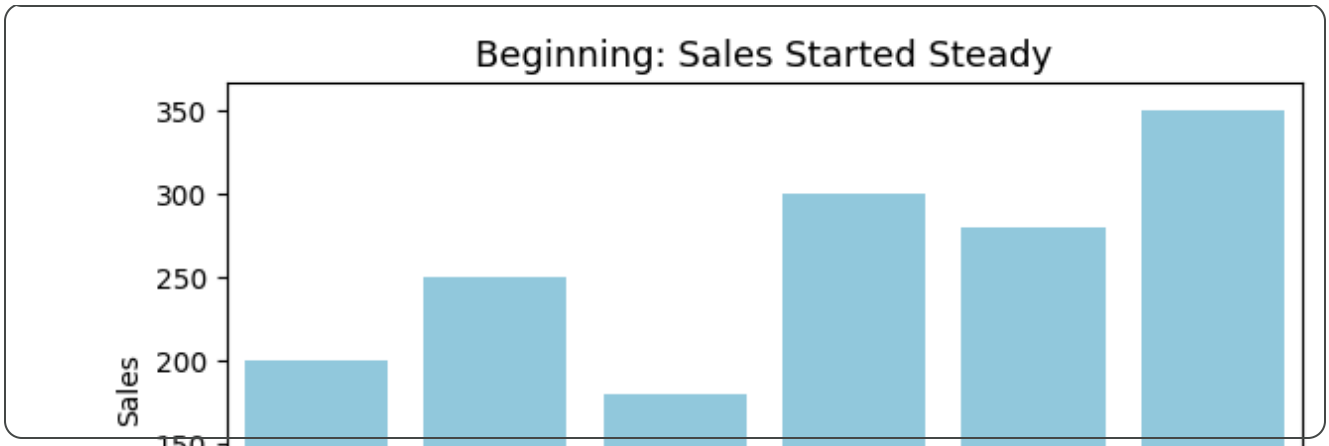


Step 4. Storytelling with Charts — Beginning, Middle, End

```
# Beginning: Context
plt.figure(figsize=(7,4))
sns.barplot(x="Month", y="Sales", data=df, color="skyblue")
plt.title("Beginning: Sales Started Steady", fontsize=13)
plt.show()

# Middle: Change or Challenge
plt.figure(figsize=(7,4))
sns.lineplot(x="Month", y="Sales", data=df, color="orange", marker="o")
plt.title("Middle: A Drop in March Due to Low Marketing", fontsize=13)
plt.show()

# End: Resolution
plt.figure(figsize=(7,4))
sns.barplot(x="Month", y="Sales", data=df, color="limegreen")
plt.title("End: Recovery and Growth After March", fontsize=13)
plt.show()
```



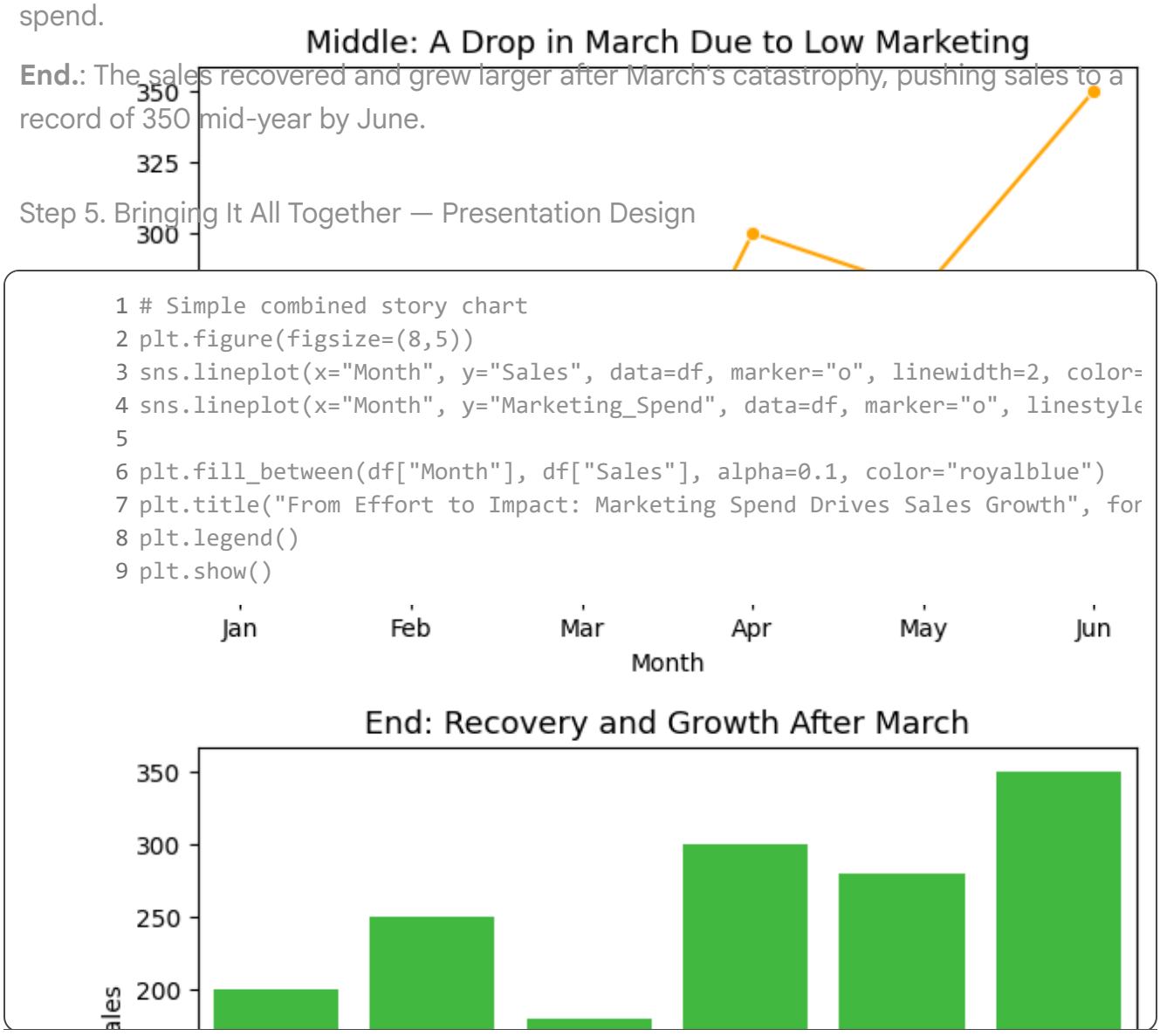
Task 3: Write one or two lines describing your data story from beginning → middle → end.

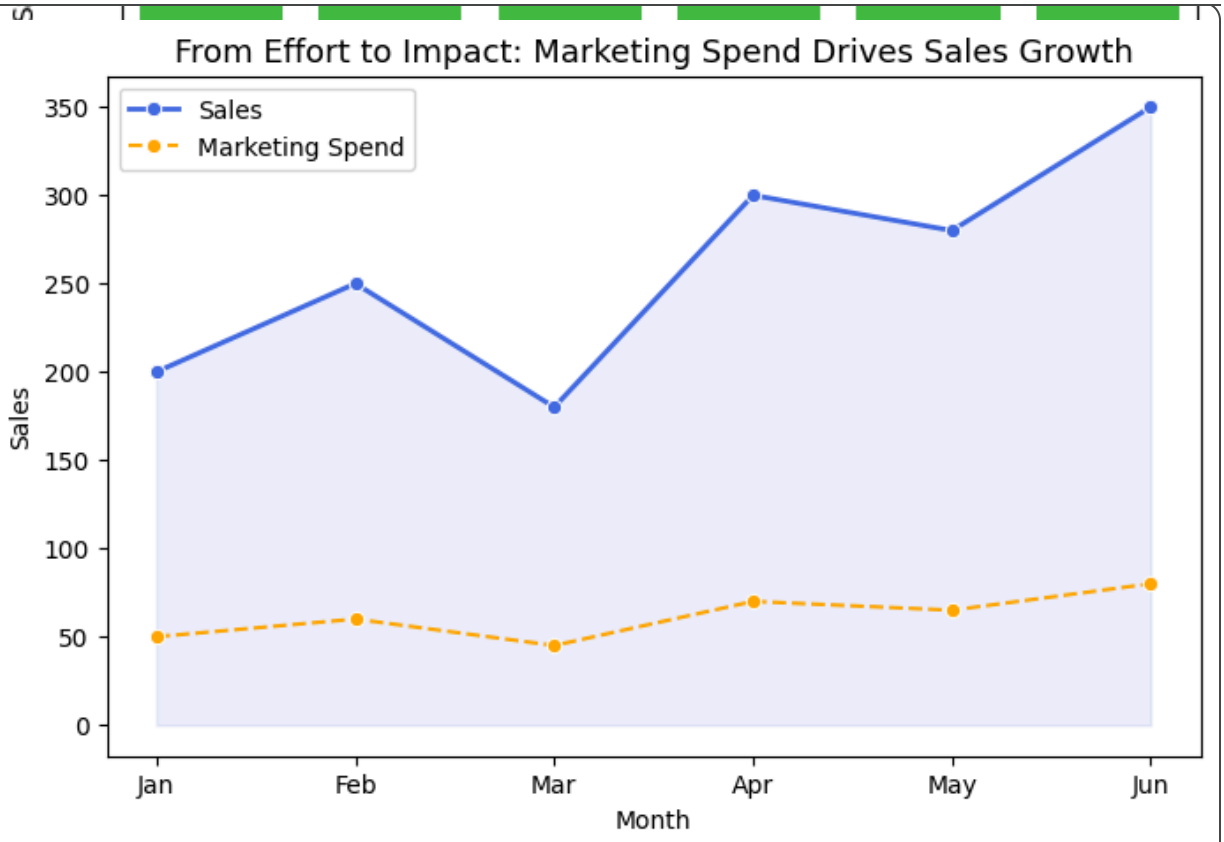
Beginning: The sales started with a steady momentum in the early year, showing an increase from January to February.

Middle: Suddenly, sales had dropped to an all-time low in March due to low marketing spend.

End.: The sales recovered and grew larger after March's catastrophy, pushing sales to a record of 350 mid-year by June.

Step 5. Bringing It All Together — Presentation Design





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5. Supplementary Activity:

1. Use a small dataset available online (kaggle, google dataset, etc.).
2. Visualize it using a bar chart.
3. Highlight one important data point (e.g., the highest or lowest value) using a distinct color.
4. Add a title and annotation that tell the story behind the data.

```
dataset = "https://raw.githubusercontent.com/Pasioncpe/CPE-031-CPE21S1---Vi

df = pd.read_csv(dataset)

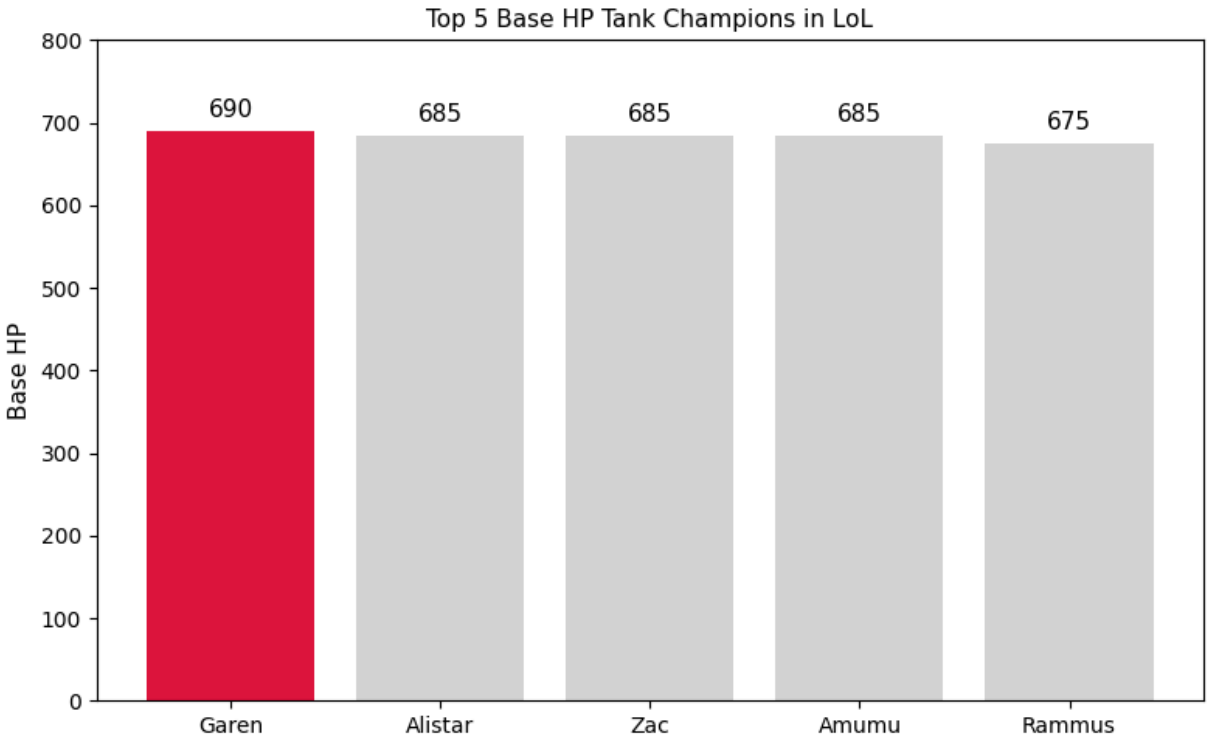
top_5_tanks = (
    df[df["Tags"].str.contains("Tank", case=False, na=False)]
    .sort_values(by=["Base HP", "HP per lvl"], ascending=[False, False])
    .head(5)
    .reset_index(drop=True)
)

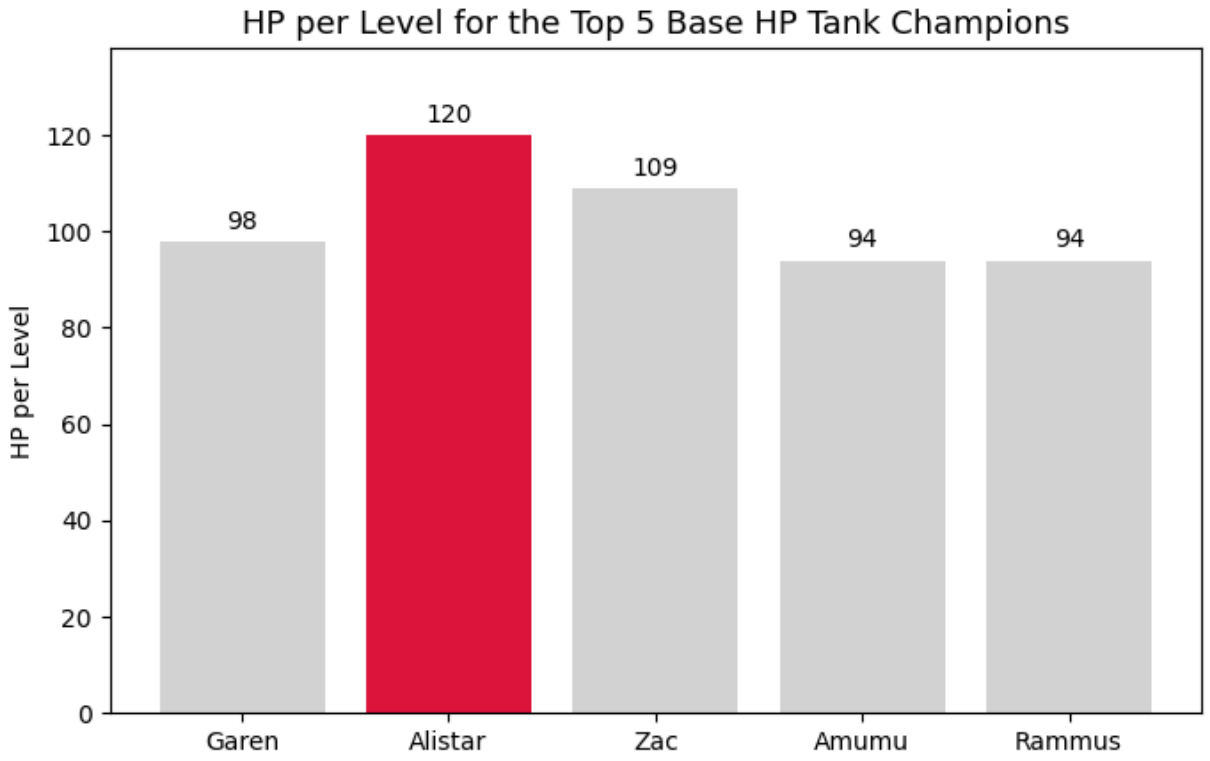
result = top_5_tanks[["Name", "Tags", "Base HP", "HP per lvl"]]
```

```
13
14 print(result.to_string(index=False))
```

```
    Name         Tags  Base HP  HP per lvl
   Garen Fighter,Tank      690          98
 Alistar Tank,Support      685         120
     Zac Tank,Fighter      685         109
   Amumu Tank,Support      685          94
  Rammus         Tank      675          94
```

```
 1 max_hp = top_5_tanks["Base HP"].max()
 2 colors = ["crimson" if hp == max_hp else "lightgray" for hp in top_5_tanks[
 3
 4 plt.figure(figsize=(8, 5))
 5 bars = plt.bar(top_5_tanks["Name"], top_5_tanks["Base HP"], color=colors)
 6
 7 plt.bar_label(bars, padding=4, fontsize=11)
 8
 9 plt.ylim(0, 800)
10
11 plt.ylabel("Base HP", fontsize=11)
12 plt.title("Top 5 Base HP Tank Champions in LoL", fontsize=11)
13 plt.tight_layout()
14 plt.show()
15
16 max_hp_lvl = top_5_tanks["HP per lvl"].max()
17 colors = ["crimson" if val == max_hp_lvl else "lightgray" for val in top_5_
18
19 plt.figure(figsize=(7, 4.5))
20 bars = plt.bar(top_5_tanks["Name"], top_5_tanks["HP per lvl"], color=colors
21 plt.bar_label(bars, padding=3)
22 plt.ylim(0, max_hp_lvl * 1.15)
23
24 plt.ylabel("HP per Level")
25 plt.title("HP per Level for the Top 5 Base HP Tank Champions", fontsize=13)
26 plt.tight_layout()
27 plt.show()
```


Top 5 Base HP Tank Champions in LoL
Base HP
800
700
600
500
400
300
200
100
0
690
685
685
685
675
Garen
Alistar
Zac
Amumu
Rammus


HP per Level for the Top 5 Base HP Tank Champions
HP per Level
120
100
80
60
40
20
0
98
120
109
94
94
Garen
Alistar
Zac
Amumu
Rammus

CHOOSING A CHAMPION

You are in a pro-level team in the game League of Legends. Your main role is a tank. For your next game, the opponents picked champions that specializes in early dives and snowballing (snowballing; a term used to describe getting an early lead by getting kills first).

You thought to yourself, **"They're definitely coming for my tower before I hit level 6, so they can get an early lead. I better prioritize my surivival early game rather than late game or else they'll get ahead. So choosing the highest Hp per Lvl will be a bad idea..."**. With this in mind, you look at your graph and see that Garen has the highest base hp. This can prevent you from giving them free kills in the early game, unabling them to snowball.

The game starts and your thought from earlier was right. At level 1, three of the enemies went straight to you to tower dive, but you choosing Garen got you the edge in